Full results for ablation study

Metric: Cityblock (L1, Manhattan)

Objective function values

					Deviation from optimal objective (%)							
Dataset	n	d	Optimal objective	UB gap (%)	GRB	rABM	ABM	ABM-i5	ABM-i10	AC	GRD	RND
Synthetic-100-2	100	2	992.92	0.00	0.00	0.58	0.00	0.00	0.00	0.00	0.00	32.38
Synthetic-100-5	100	5	2,162.49	-0.34	0.00	0.89	0.00	0.00	0.00	0.95	10.19	28.50
Synthetic-100-10	100	10	4,485.43	0.00	0.00	1.54	0.33	0.09	0.09	0.55	0.00	27.28
Synthetic-1000-2	1,000	2	9,954.08	0.00	0.00	0.14	0.02	0.00	0.00	0.04	0.13	34.61
Synthetic-1000-5	1,000	5	21,619.29	-0.42	0.00	0.21	0.05	0.00	0.00	0.66	9.71	31.40
Synthetic-1000-10	1,000	10	44,851.80	-0.01	0.00	0.19	0.07	0.01	0.00	0.53	0.00	27.86
Synthetic-2000-2	2,000	2	19,911.24	0.00	0.00	0.13	0.00	0.00	0.00	0.05	0.21	35.74
Synthetic-2000-5	2,000	5	43,236.79	-0.36	0.00	0.33	0.05	0.00	0.00	0.63	9.87	30.71
Synthetic-2000-10	2,000	10	89,728.14	-0.01	0.00	0.28	0.08	0.01	0.00	0.54	0.00	28.52
Synthetic-3000-2	3,000	2	29,878.95	0.00	0.00	0.02	0.02	0.00	0.00	0.04	0.17	34.71
Synthetic-3000-5	3,000	5	64,854.60	-0.36	0.00	0.20	0.05	0.00	0.00	0.57	9.83	30.64
Synthetic-3000-10	3,000	10	134,560.63	-0.01	0.00	0.20	0.06	0.00	0.00	0.53	0.00	28.72
Average					0.00	0.39	0.06	0.01	0.01	0.43	3.34	30.92

Runtimes (seconds)

			Runtime (s)							
Dataset	n	d	GRB	rABM	ABM	ABM-i5	ABM-i10	AC	GRD	RND
Synthetic-100-2	100	2	0.02	<0.01	<0.01	0.08	0.16	<0.01	<0.01	<0.01
Synthetic-100-5	100	5	0.01	<0.01	<0.01	0.08	0.16	<0.01	<0.01	<0.01
Synthetic-100-10	100	10	0.01	<0.01	<0.01	0.08	0.16	<0.01	<0.01	<0.01
Synthetic-1000-2	1,000	2	1.02	0.03	0.03	0.26	0.49	0.40	0.38	<0.01
Synthetic-1000-5	1,000	5	1.26	0.03	0.02	0.18	0.33	0.40	0.38	<0.01
Synthetic-1000-10	1,000	10	0.99	0.02	0.02	0.18	0.33	0.40	0.38	<0.01
Synthetic-2000-2	2,000	2	4.14	0.24	0.18	1.43	2.75	2.70	1.74	<0.01
Synthetic-2000-5	2,000	5	5.03	0.19	0.14	0.99	1.86	2.69	1.73	<0.01
Synthetic-2000-10	2,000	10	4.01	0.15	0.10	0.80	1.50	2.70	1.71	<0.01
Synthetic-3000-2	3,000	2	9.95	0.73	0.62	4.45	8.29	8.72	4.20	<0.01
Synthetic-3000-5	3,000	5	12.27	0.58	0.41	3.00	5.61	8.73	4.24	<0.01
Synthetic-3000-10	3,000	10	9.92	0.47	0.31	2.40	4.43	8.73	4.15	<0.01
Average			4.05	0.20	0.15	1.16	2.17	2.96	1.58	<0.01

Metric: Euclidean (L2)

Objective function values

					Deviation from optimal objective (%)							
Dataset	n	d	Optimal objective	UB gap (%)	GRB	rABM	ABM	ABM-i5	ABM-i10	AC	GRD	RND
Synthetic-100-2	100	2	746.75	-0.43	0.00	0.96	0.00	0.00	0.00	0.20	3.83	30.58
Synthetic-100-5	100	5	1,062.90	-1.47	0.00	0.35	0.00	0.00	0.00	0.84	8.08	23.38
Synthetic-100-10	100	10	1,606.27	-0.40	0.00	0.43	0.07	0.01	0.00	0.23	0.00	23.78
Synthetic-1000-2	1,000	2	7,486.37	-0.51	0.00	0.21	0.03	0.00	0.00	0.26	3.98	32.51
Synthetic-1000-5	1,000	5	10,632.73	-1.66	0.00	0.11	0.02	0.00	0.00	0.86	7.92	26.52
Synthetic-1000-10	1,000	10	16,071.76	-0.48	0.00	0.08	0.02	0.00	0.00	0.35	0.01	24.67
Synthetic-2000-2	2,000	2	14,970.11	-0.56	0.00	0.17	0.13	0.00	0.00	0.20	3.94	33.61
Synthetic-2000-5	2,000	5	21,259.64	-1.64	0.00	0.17	0.01	0.00	0.00	0.99	8.11	25.67
Synthetic-2000-10	2,000	10	32,151.88	-0.42	0.00	0.08	0.01	0.00	0.00	0.35	0.01	25.31
Synthetic-3000-2	3,000	2	22,463.99	-0.55	0.00	0.06	0.04	0.00	0.00	0.18	3.94	32.60
Synthetic-3000-5	3,000	5	31,894.50	-1.61	0.00	0.11	0.00	0.00	0.00	1.00	7.88	25.67
Synthetic-3000-10	3,000	10	48,218.94	-0.51	0.00	0.06	0.07	0.00	0.00	0.31	0.01	25.45
Average					0.00	0.23	0.03	0.00	0.00	0.48	3.98	27.48

Runtimes (seconds)

			Runtime (s)							
Dataset	n	d	GRB	rABM	ABM	ABM-i5	ABM-i10	AC	GRD	RND
Synthetic-100-2	100	2	0.01	<0.01	<0.01	0.08	0.16	<0.01	<0.01	<0.01
Synthetic-100-5	100	5	0.01	<0.01	<0.01	0.08	0.16	<0.01	<0.01	<0.01
Synthetic-100-10	100	10	0.01	<0.01	<0.01	0.08	0.16	<0.01	<0.01	<0.01
Synthetic-1000-2	1,000	2	1.40	0.05	0.03	0.26	0.49	0.40	0.39	<0.01
Synthetic-1000-5	1,000	5	1.48	0.04	0.03	0.22	0.43	0.40	0.38	<0.01
Synthetic-1000-10	1,000	10	1.28	0.03	0.02	0.18	0.33	0.40	0.38	<0.01
Synthetic-2000-2	2,000	2	5.94	0.28	0.18	1.40	2.63	2.69	1.74	<0.01
Synthetic-2000-5	2,000	5	6.74	0.22	0.18	1.17	2.16	2.69	1.74	<0.01
Synthetic-2000-10	2,000	10	5.89	0.16	0.11	0.90	1.65	2.69	1.71	<0.01
Synthetic-3000-2	3,000	2	13.64	0.86	0.59	4.24	7.93	8.72	4.19	<0.01
Synthetic-3000-5	3,000	5	15.61	0.66	0.62	3.61	6.54	8.73	4.24	<0.01
Synthetic-3000-10	3,000	10	14.66	0.49	0.33	2.66	4.97	8.73	4.17	<0.01
Average			5.56	0.23	0.17	1.24	2.30	2.95	1.58	<0.01

Metric: Squared Euclidean (L2 squared)

Objective function values

					Deviation from optimal objective (%)							
Dataset	n	d	Optimal objective	UB gap (%)	GRB	rABM	ABM	ABM-i5	ABM-i10	AC	GRD	RND
Synthetic-100-2	100	2	12,046.92	-1.57	0.00	0.76	0.01	0.00	0.00	0.06	2.03	41.87
Synthetic-100-5	100	5	24,473.15	-0.19	0.00	1.28	0.02	0.00	0.00	0.57	7.34	35.11
Synthetic-100-10	100	10	52,869.55	-0.53	0.00	1.38	0.33	0.08	0.00	1.29	0.01	35.83
Synthetic-1000-2	1,000	2	121,095.14	-1.91	0.00	0.05	0.00	0.00	0.00	0.12	2.23	44.97
Synthetic-1000-5	1,000	5	245,144.11	-0.30	0.00	0.33	0.00	0.00	0.00	0.95	7.46	38.90
Synthetic-1000-10	1,000	10	529,425.22	-0.67	0.00	0.24	0.06	0.00	0.00	0.90	0.01	36.73
Synthetic-2000-2	2,000	2	242,293.64	-1.89	0.00	0.03	0.00	0.00	0.00	0.09	2.24	46.78
Synthetic-2000-5	2,000	5	489,931.39	-0.40	0.00	0.42	0.00	0.00	0.00	0.94	7.47	38.02
Synthetic-2000-10	2,000	10	1,059,365.74	-0.55	0.00	0.27	0.05	0.00	0.00	1.00	0.01	37.60
Synthetic-3000-2	3,000	2	363,489.14	-1.86	0.00	0.04	0.00	0.00	0.00	0.14	2.24	45.16
Synthetic-3000-5	3,000	5	735,153.35	-0.45	0.00	0.28	0.00	0.00	0.00	0.96	7.42	37.68
Synthetic-3000-10	3,000	10	1,588,665.49	-0.74	0.00	0.21	0.18	0.00	0.00	0.94	0.01	37.56
Average					0.00	0.44	0.05	0.01	0.00	0.66	3.20	39.68

Runtimes (seconds)

			Runtime (s)							
Dataset	n	d	GRB	rABM	ABM	ABM-i5	ABM-i10	AC	GRD	RND
Synthetic-100-2	100	2	0.12	<0.01	<0.01	0.08	0.16	0.01	<0.01	<0.01
Synthetic-100-5	100	5	0.07	<0.01	<0.01	0.08	0.16	0.01	<0.01	<0.01
Synthetic-100-10	100	10	0.02	<0.01	<0.01	0.08	0.16	0.01	<0.01	<0.01
Synthetic-1000-2	1,000	2	4.24	0.04	0.03	0.19	0.34	0.64	0.38	<0.01
Synthetic-1000-5	1,000	5	3.28	0.03	0.03	0.18	0.34	0.98	0.39	<0.01
Synthetic-1000-10	1,000	10	2.68	0.03	0.02	0.18	0.33	1.11	0.38	<0.01
Synthetic-2000-2	2,000	2	18.69	0.24	0.19	1.16	2.11	4.95	1.75	<0.01
Synthetic-2000-5	2,000	5	14.45	0.22	0.19	1.10	2.03	7.64	1.73	<0.01
Synthetic-2000-10	2,000	10	12.54	0.17	0.12	0.78	1.44	6.70	1.72	<0.01
Synthetic-3000-2	3,000	2	33.49	0.71	0.62	3.51	6.38	16.57	4.18	<0.01
Synthetic-3000-5	3,000	5	28.57	0.66	0.59	3.40	6.26	25.74	4.24	<0.01
Synthetic-3000-10	3,000	10	24.32	0.50	0.45	2.36	4.31	23.92	4.18	<0.01
Average			11.87	0.22	0.19	1.09	2.00	7.35	1.58	<0.01